

SCHOOL OF ENGINEERING, CUSAT
B. TECH DEGREE 2ND SEMESTER FIRST INTERNAL EXAMINATIONS, APRIL 2024
23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++
(CS A/B, EE-B)
(2023 Scheme)

TIME: 2 hrs**MAX. MARKS: 30****Course Outcomes**

On completion of this course the student will be able to:

CO1	Understand the basic concepts of object oriented programming
CO2	Describe the object oriented programming with concepts of streams, classes, functions, data and objects
CO3	Implement object oriented programming construct like encapsulation, inheritance and polymorphism.
CO4	Understand dynamic memory management technique using pointers, constructors, destructors
CO5	Identify classes including data, methods and the relationship among the classes from a given statement and solve problem using object oriented constructs in C++.

PART A**(Answer all questions)****(5 x 2= 10 Marks)**

	Questions	Marks	BL	CO	PO
I (a)	Explain the use of 'this' pointer.	2	L1	1	1
(b)	Differentiate between Constructor and Destructor	2	L2	4	3,1,2
(c)	What is object? How they are created.	2	L1	2	1,2
(d)	Explain static data member with an example.	2	L1	2	1,2
(e)	What is the use of new and delete operators?	2	L1	4	3,1,2

PART B

**(Answer the following questions
choosing either the first OR second option)**

(1 x 10= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Explain basis concepts of OOPs.	5	L1	3	1,2,3
(b)	Write a C++ program to print and find the sum of 20 prime numbers.	5	L3	2	1,2
OR					
III (a)	Difference between procedural oriented programming and Object oriented programming.	5	L1	3	1,2,3
(b)	Write a C++ program to print range of prime numbers.	5	L2	2	1,2
(1 x 10= 10 Marks)					
IV (a)	Write a program to print the names of student by creating a student class. If no name is passed while creating an object of the student class, then the name should be "unknown", otherwise the name should be equal to the string value passed while creating the object.	5	L3	4	3,1,2
(b)	Explain in detail about friend function.	5	L1	4	3,1,2

OR

V (a)	Write a C++ program to use friend function acting as bridge between two classes (Sum of internal mark and university marks stored of a student stored in two separate classes for 5 subjects). Prepare a mark list for the student.	5	L3	4	3,12
(b)	Predict the output and explain it. <pre> #include <iostream.h> using namespace std; class A { int x; public: A(int d) { x=d; } ~A() { cout<<x; } }; int main() { A a(50),b(60); for(int i=1;i<=10;i++) { A y(i); } A c(70); for(int i=11;i<=20;i++) { A z(i); } return 0; } </pre>	5	L3	4	3,12